Problem Chosen	2023 MCM/ICM Summary Sheet	Team Control Number
Y		2330740

Regress To All Blue

Summary

As sailing became a popular form of recreation and competitive sport, the used sailboat market is thriving. Entrusted by a sailing broker from Hong Kong, our team built a mathematical model to generate a second-hand sailing pricing report, hoping to provide some reference for his sailing pricing.

For problem 1, we comprehensively used **Multiple Linear Regression** and **Random Forest Regression** to fit the data, hoping to get a more accurate fitting curve. Through the interpretation of the regression equation, the influence direction and degree of the core explanatory variable on the dependent variable Price are found. Random forest regression also concluded that the characteristics of sailboat variants are the main factors affecting the pricing.

In problem 2, we adopted the method of **Cross-Variable Consistency Test** and **Cross-Introduction** of variable parameters to further analyze the influence direction of specific regional effects on the prices of different sailing varieties. First of all, we noticed that the variant's characteristic variables is too much, to classify it as a variable calculation process would be too multifarious, so we use factor analysis method under conditions allowing for three indicators dimension reduction. Using the two models in the first question, we can naturally explain the influence of regional effect on Price. Then, by using the cross-variable consistency test and the cross-introduction of variable parameters, we get the conclusion that the influence of the regional effect on different variants is not completely consistent.

In terms of problem3, we have mainly applied the results of the previous two questions to the data for Hong Kong to see if there is a correlation. Through the analysis of the calculation results, the feasibility of our model is confirmed, and Hong Kong is analyzed independently.

Finally,we optimize random forest model based on **Hyper Parameter Optimization**. The specific method is through **Randomized Search on hyper parameters** and **Grid Search**. Our two models can complement each other and enhance the the accuracy of the results. Through optimization of Random Forest Regression, the accuracy of the model will be further improved.

Keywords: Multiple Linear Regression, Random Forest Algorithm, Cross-Variable Consistency Test, Hyper Parameter Optimization

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1 Introduction

1.1 Restatement of the Problem

As sailing became a popular form of recreation and competitive sport, the demand for sailboats has increased. Sailboats hold their value well, making them a good investment, and many enthusiasts prefer to purchase second-hand boats for affordability. However, the value of sailboats varies with age and market conditions, and there is a lack of objective pricing standards.

In an effort to better understand the sailboat market, a sailboat broker in Hong Kong has commissioned our team to study the pricing of used sailboats. Since the price is bound to be affected by various factors, used sailboats price evaluation model is challenging but worthy of effort.

The model we are to built should solve the following problems:

- Problem 1: Develop a mathematical model to explain sailboat listing prices using relevant predictors. Discuss the precision of each sailboat variant's price estimate.
- Problem 2: Analyze the effect of region on prices and discuss if it's consistent across all sailboat variants. Explain its practical and statistical significance.
- Problem 3: Discusses the expansion and adaptation performance of regional influences in the model used in Hong Kong. Use models to explore what regional effects Hong Kong has on different sailing subsets and whether the effects are the same.

1.2 Our work

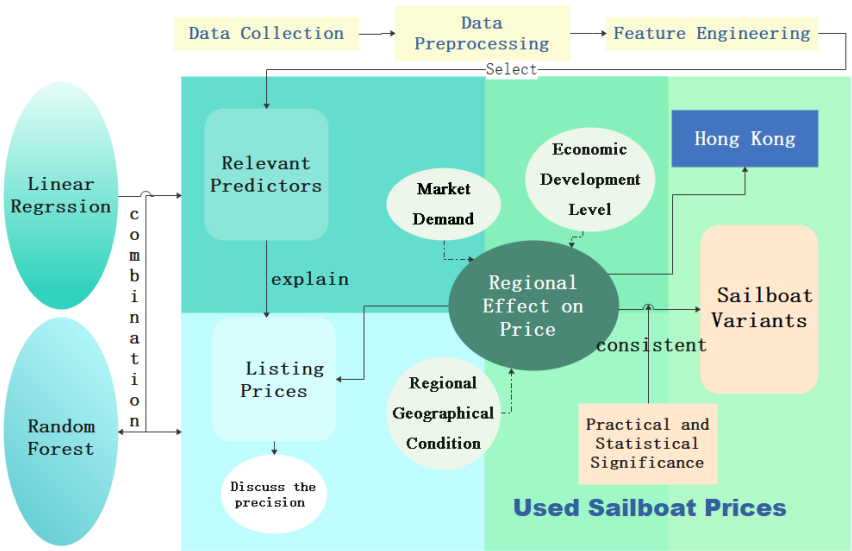


Figure 1: Our Work

2 Assumptions and Justification

- **Assumption 1 :** The sample data source is authentic and the data is highly accurate.
  Justification: In addition to the official data set provided, our data are collected from various major sailing websites, as well as geographical and economic information from national official websites. The data are of good authenticity and accuracy.

- **Assumption 2 :** Used sailboat trading is not subject to price manipulation by third-party platforms.
  Justification: We assume that there is no platform control over prices and excessive commission, resulting in excessive interference with prices. The second-hand car trading website is a medium for trading between buyers and sellers, serving as a platform. It is reasonable to assume that there is no black box operation.

- **Assumption 3 :** Used sailboat sellers do not over-subjective pricing, pricing according to market rules, can be interpreted and predicted.
  Justification: Second-hand sailboat sellers sell sailboats for profit and do not overprice them. In a free market where supply and demand dominate, prices can be influenced and explained by supply and demand.

- **Assumption 4 :** The transaction price of second-hand cars is not much affected by the buyer's personal preferences.
  Justification: There is no black Swan event and no major natural disaster such as tsunami.

- **Assumption 5 :** The transaction price of second-hand cars is not much affected by the buyer's personal preferences.
  Justification: In case of a major accident, the price of second-hand ships will inevitably fluctuate greatly, but this is not caused by the core explanatory variable and has nothing to do with the study in this paper.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations Table

Notations	Description
P_k	the price when sample is k
X_i	quantitative features
A_j B_j C_j	classification feature
β_i β_{ij}	coefficient
ε	perturbation term
I_i	interaction terms
J_i	interaction terms

4 Interpretation Model based on MLR and Random Forest Algorithm

4.1 Data Preprocessing

Due to the limited nature of the data set, it is necessary to supplement the data with additional resources. Before data analysis, the availability of data must be guaranteed.

• **Supplement data set** by collecting data from various websites, such as SailboatData, BoatTrader, SailboatListings, World Sailing, United States Geological Survey, etc., as the price is bound to be affected by factors other than the dataset given in the title.

• **Missing value processing** by supplementing sailing variety parameters through data search, and deleting other missing values due to their small number.

• **Data standardization** by previewing its distribution and processing some of it logarithmically to conform to normal distribution as much as possible, as variables need to roughly conform to normal distribution for fitting in the multiple linear regression model.

• **Outlier identification and processing** by detecting them based on the three principles and deleting abnormal data according to the actual situation as the data distribution approximates geometric distribution.

4.2 Parameters Selection

Used sailboats, like many luxury items, are influenced by a range of factors that fall into three categories: sailboat characteristics, regional factors, and market conditions.

Sailboat features include the make and variant, structural parameters, interior configuration, age, and newness of the sailboat. Regional factors depend on location, environmental factors, and culture. Coastal cities and areas with less severe weather tend to have a greater demand for sailboats. Market conditions take into account the economic conditions and per capita income of the region during a specific period.

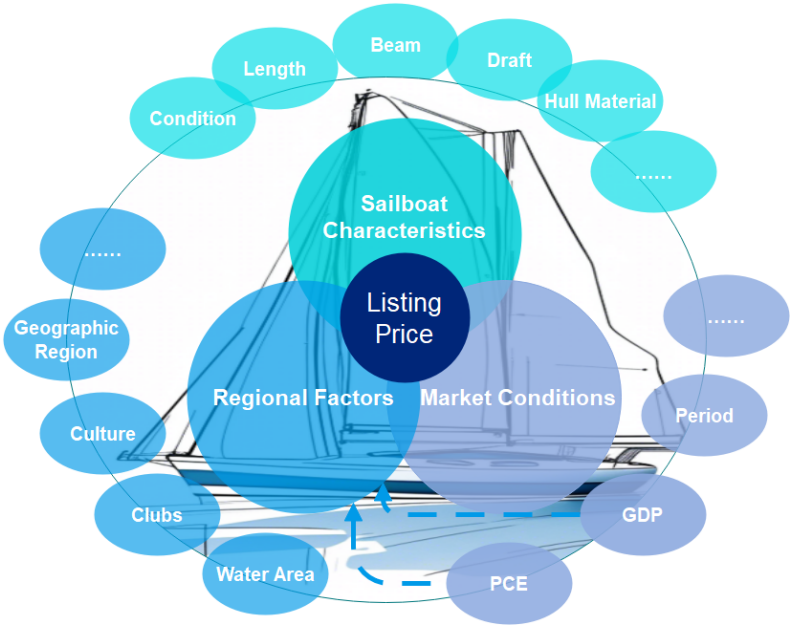


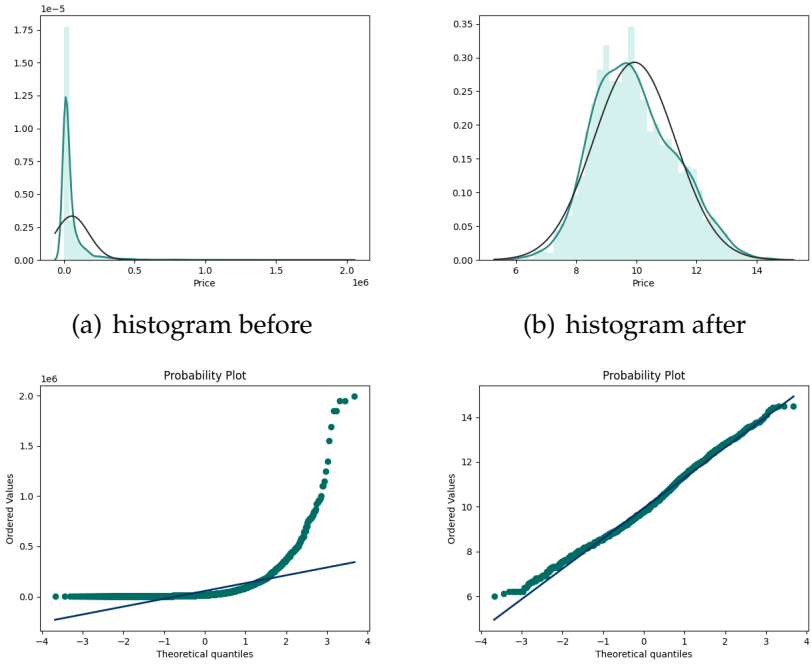
Figure 2: Influence Factors

With these factors in mind, we collected data parameters as shown in the table below from different sources to further refine our analysis. Given the complexity of the relationship between the factors, we need to select indicators that are best suited to model establishment.

Table 2: Notations Table

Features	Features	Features	Features
Make	Variant	Motor Type	Hull Material
Variant	PCE	Displacement	Engine Count
Length	GDP	Project boat	Geographic Region
Beam	Age	Is Diesel	Country/Region/State
Draft	Condition	Is Mono	Clubs Number
Type	Sail Area	Water Area	Cabins Number

To do this, first transform categorical features using Target Encoding. We then perform a logarithmic transformation of the quantitative data so that it roughly follows a normal distribution in favor of eliminating the effects of heteroscedasticity. We do not process discontinuous features with size relationships.



(a) histogram before (b) histogram after

(c) quantile-quantile plot before (d) quantile-quantile plot after

Figure 3: Price Logarithmic Conversion Comparison

Next, explore the correlation between different features and prices. For quantitative data, we plot the correlation heat map using the Pearson correlation coefficient, set the correlation threshold to 0.1, and remove features with less correlation. [3] For categorical features, we use independent sample t-tests and ANOVA to explore correlation with prices according to the number of categories.

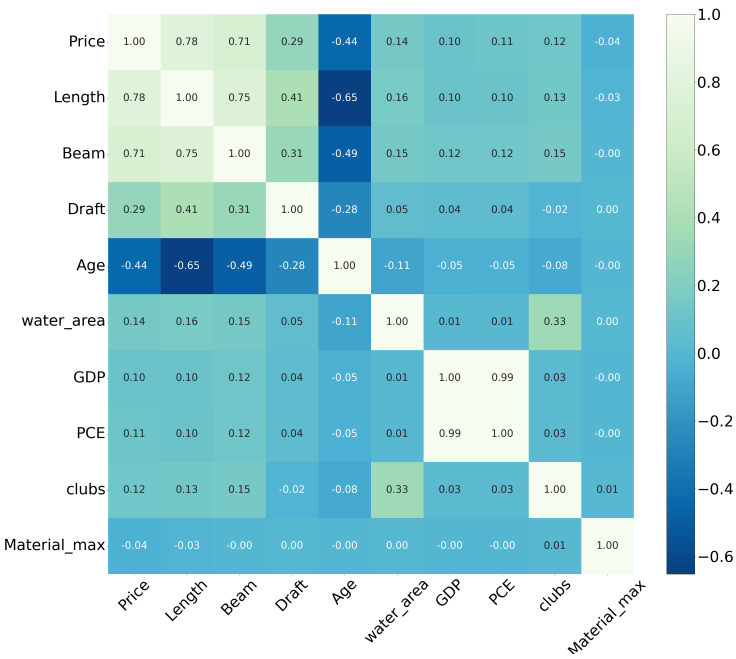


Figure 4: Correlation coefficient heat map

Finally, according to the correlation filter, the last independent variables left are

length, beam, draft, type, engine count, age, condition, mono, diesel, project, water area, clubs number, PCE, GDP.

4.3 Establishment of Model

4.3.1 Multiple Linear Regression Model

Regression analysis is an extremely powerful tool for identifying statistical relationships between pricing and multiple complex explanatory variables. [5] Take the following steps to establish a multiple linear regression equation using Ordinary Least Square method.

1. Set variables

Choose quantitative data as independent variables. Transform multicategorical variables into dummy variables.

The regression model form we seted as follow:

$$P_k = \beta_0 + \beta_1 X_1 + \cdots + \beta_{10} X_{10} + \beta_{1j} A_j + \beta_{2j} B_j + \beta_{3j} C_j + \varepsilon \tag{1}$$

i=1,2...10. j depends on the number of categorical variables.

where P_k is the response that corresponds to the levels of the explanatory variables X_i, A_j, B_j, C_j at kth observation. β_i and β_{ij} are the coefficients in the linear relationship, ε is perturbation term that create scatter around the linear relationship. Because there are so many variables, they are displayed in the figure below.

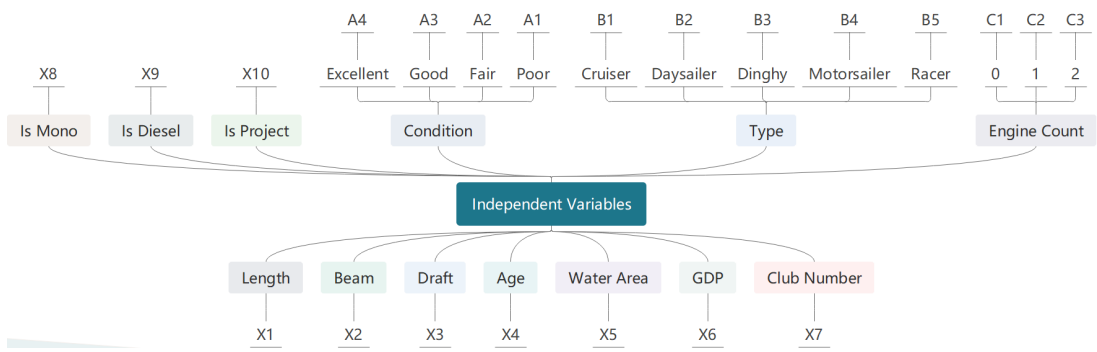


Figure 5: Variables Set

In addition, we ensured that each variable was properly formatted.

2. Model Suitability Test

Prob>F = 0.0000 <0.05, which means that the original hypothesis is rejected at a 95% confidence level, indicating that the model has passed the joint significance test and is therefore meaningful. This means that we can be confident in the results of our model and that it accurately represents the relationships between the variables.

3. Heteroscedasticity Test

Use White test to detect heteroscedasticity and reject the null hypothesis. Apply OLS and Heteroskedasticity-Robust Standard Error to address it. This step ensures that our model is not biased due to unequal variances across the independent variables.

4. Multicollinearity Test

In Regression analysis, there is a tendency for independent variables to correlate with each other, which is called Multicollinearity. The variance inflation factor (VIF) is

used to detect multicollinearity. The higher the VIF, the worse the Multicollinearity. It is generally considered that when VIF is greater than 10, the representative model has serious collinearity problem.

$$VIF_m = \frac{1}{1 - R^2} \tag{2}$$

where k is independent variables numbers, R^2 is the goodness of fit obtained by regressing the mth independent variable as the dependent variable on the remaining k-1 independent variables.

It is discovered that there is multicollinearity between GDP and PCE, and the feature PCE is removed. In other words, we eliminated any redundant information to ensure that our model includes only unique and useful data.

Table 3: Model Estimatation Results

Variable	VIF before	VIF after
PCE	43.4	-
GDP	43.37	1.03

5. Multiple Regression Equation Solution

Use the adjusted data to calculate the regression coefficient and derive the regression equation. We then analyzed the results to gain insights into the relationships between the variables and to make predictions and explanations based on the model. The following multiple linear regression equation was obtained by calculation:

$$\begin{aligned} P_k = &-4.489 + 3.268X_1 + 0.769X_2 - 0.068X_3 + 0.008X_4 \\ &-0.012X_5 + 0.166X_6 + 0.0002X_7 + 0.463X_8 - 0.368X_9 \\ &+1.633X_{10} - 1.222A_2 - 0.725A_3 - 0.102A_4 + 0.016B_1 \\ &+0.217B_2 + 1.121B_3 - 0.03B_5 - 0.101C_2 + 0.158C_3 \end{aligned} \tag{3}$$

6. Goodness-of-fit Test

Use Adjusted R-Squared to test the goodness of fit.

$$\begin{aligned} SSE &= \sum_{i=1}^{n} (\hat{y} - y_i)^2 \\ SST &= \sum_{i=1}^{n} (y_i - \bar{y})^2 \\ adjustedR^2 &= 1 - \frac{SSE(n-1)}{SST(n-k-1)} \end{aligned} \tag{4}$$

where k is the number of independent variables, n is the number of data.

It provides an indication of how well the model fits the data. Overall, our model has a good fit and is able to explain a significant amount of the variation in the dependent variable.

4.3.2 Random Forest Regression Model

The random forest algorithm is based on a decision tree algorithm. [2] There are two types of random forest algorithms: regression and classification. [1] In the random

forest regression, the output is in numerical or continuous manner. For this study, it is reasonable to use random forest regression. By using this method, we were able to gain valuable insights into the relationships between different features and how they contribute to the overall prediction.

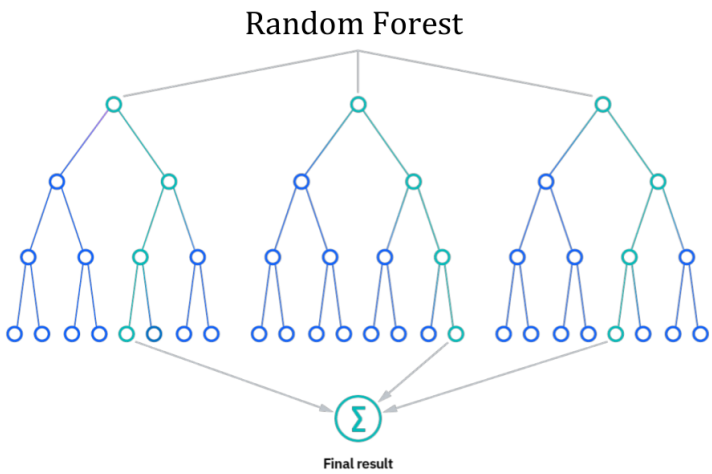


Figure 6: Random Forest

In order to develop an accurate predictive model, we utilized the random forest regression algorithm for fitting. This method allowed us to effectively consider a multitude of factors and variables that could impact the outcome. Additionally, we carefully calculated the model parameters and evaluated the importance of each feature to ensure the highest level of accuracy and reliability.

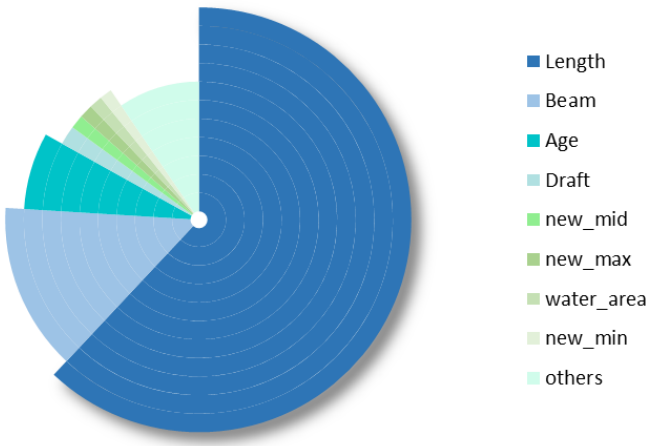


Figure 7: Features Importance Score

To further enhance the accuracy of model, select the first 35 features for k-Fold Cross Validation with k=10. This approach allowed all data participate in training and prediction, effectively avoiding overfitting, fully embodying the idea of crossover. Ensure model remains robust and reliable.

After conducting the K-fold cross validation [4], we obtained ten prediction scores, which were then averaged to obtain the final score of the model. This approach effectively evaluate the overall performance of the model and ensure that it is capable of accurately predicting outcomes across a range of different scenarios and contexts.

4.3.3 Model Result

Used sailboat price regression explanatory model has been analyzed with three performance parameters i.e. Adjusted R^2 score, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Out of the two regression models, random forest regression model has performed better with values 0.8326 as Adjusted R^2 Score, 0.3902 as MAE and 0.5369 as RMSE.

Table 4: Model Estimatation Results

Model	**MAE**	**RMSE**	**Adjusted R^2**
Linear Regrssion	0.4215	0.6689	0.7596
Random Forest	0.3902	0.5369	0.8326

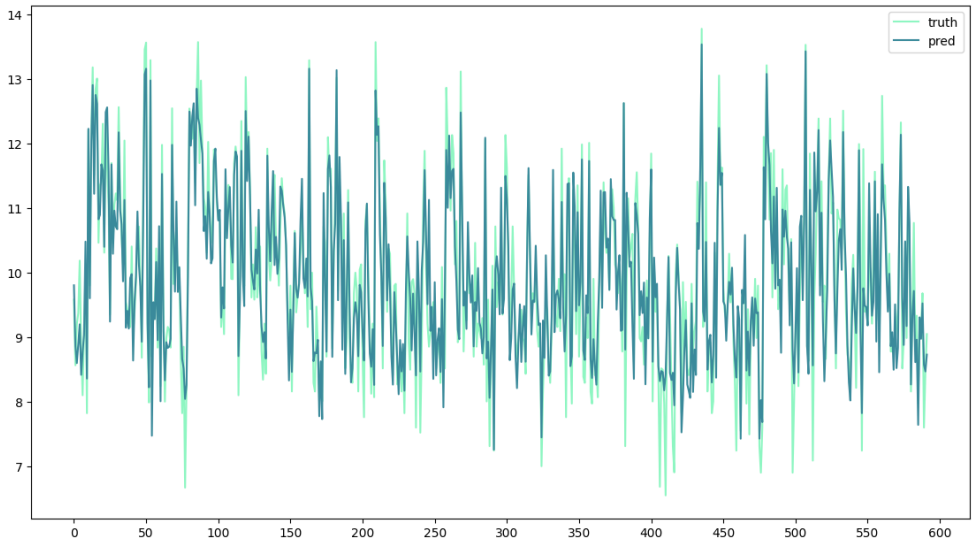


Figure 8: Model Predictions and Actual values Comparison

In the models of predicting and explaining commodity pricing, multiple linear regression model and random forest regression are the two most commonly used models. It can be seen from the figure below that the two can complement each other and enhance the accuracy of the results.

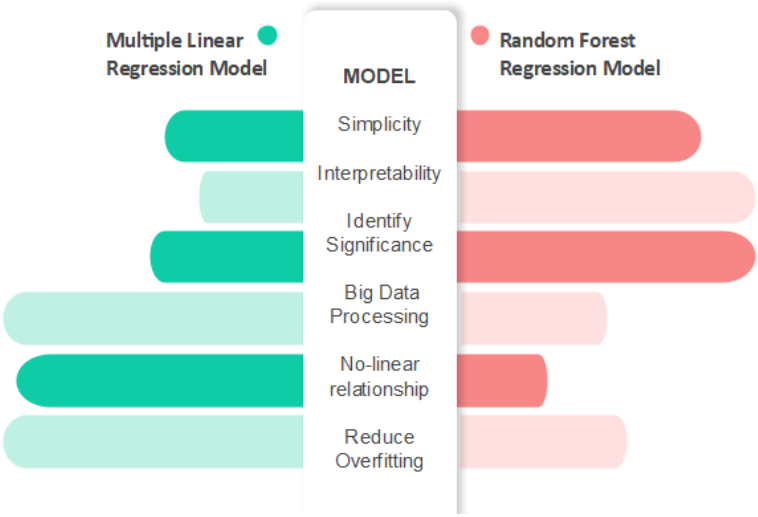


Figure 9: Model Comparison

Part of the test set is input into the random forest regression model, and we randomly select 12 sailing brands, and the difference and error between the predicted price and the actual price are shown in the figure.

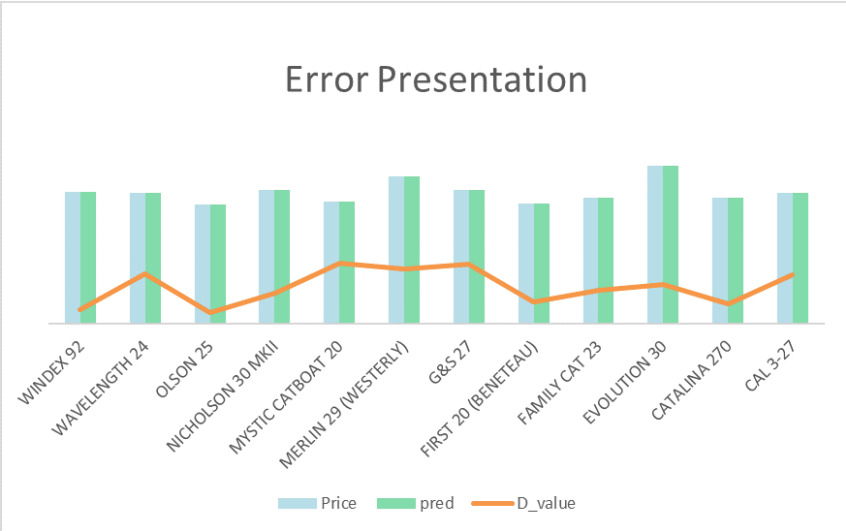


Figure 10: Model Result

5 Testing Model based on Interaction Factor Consistency Test

5.1 Interaction Factor Consistency Test

Introduce GDP, PCE, Club Number and Water Area as regional factors, and now discuss the influence of these factors on the price in the regression model.

• For the PCE variable, in order to eliminate the influence of multicollinearity, it is removed from the independent variable, and its impact on the price cannot be judged, which is a necessary sacrifice to maintain the stability of the model. From a practical

point of view, PCE and GDP are generally positively correlated, and the regional effects of the two are homotropic.

• For GDP variable, the regression coefficient β_6>0 is different from 0 at 95% confidence level, which means that GDP has a positive effect on Price when other variables are certain.

• For Club Number, the regression coefficient β_7>0 is different from 0 at 90% confidence level, which means that Club Number has a positive effect on Price when other variables are certain.

• For the variable Water Area, the regression coefficient β_5<0 and is different from 0 at 90% confidence level, which means that Club Number has a negative effect on Price when other variables are certain.

The regional effect is classified into regional economic development level effect, regional geographical condition effect and regional market demand effect based on three characteristic variables: GDP, Water Area, Club Number.

Due to the excessive number of sailboat varieties in the data set, the structure and characteristics of specific varieties are unique. Using its combination of characteristic indicators to refer to sailboat's variant, and putting them as new elements into the model to discuss whether regional impacts are consistent across different sailing boat variants.

Firstly, the three classification indicators are not easy to solve, and they also have to consider their correspondence with variants, which undoubtedly greatly increases the difficulty of data processing. Naturally, we thought of dimensionality reduction of variable indicators, and through the KMO test, it was found that the dimensionality of three indicators could be reduced using factor analysis without losing the main characteristics. we use factor analysis to reduce the dimensionality of the characteristic variables of sailboat varieties, and perform kmo and bartlett tests on the three variables: Length, Beam, Draft.

Do KMO and Bartlett's Test and result as follow:

Table 5: KMO and Bartlett's Test

KMO and Bartlett's Test		
KMO Measure		0.684
Bartlett's Test of Sphericity	Approx. Chi-Square	5946.116
	df	3
	Sig.	0.000***

The results were KMO>0.6 and Sig.<0.05, allowing for factor analysis. The following figure is a load matrix heat map, which can analyze the importance of hidden variables in each factor. The lighter the color of the heat map, the greater the correlation.

Figure 11: Factor Load Matrix Heat Map

By doing factor analysis, get the following dimensionally reduced variable C.

$$C = 0.453X_1 + 0.435X_2 + 0.32X_3 \tag{5}$$

where X_1, X_2, X_3 is Length, Beam, Draft, respectively.

Next, we performed interaction term consistency analysis. Add interaction termsI_1, I_2, I_3 replace the independent variable X_1, X_2, X_3 with C, and incorporate the three interaction terms into the model.

$$\begin{aligned} I_1 &= C * GDP \\ I_2 &= C * ClubNumber \\ I_3 &= C * WaterArea \end{aligned} \tag{6}$$

Then, we constructed t-statistic for consistency tests to obtain the P-test values for each indicator.

$$t = \frac{\hat{\beta}_i}{St.Err.} \sim t(n-k-1) \tag{7}$$

where k is the number of independent variables, n is the number of data, St. Err is standard error.

Analyzing the above table, it can be seen that C, Water Area, GDP, Club Number alone all have an impact on the price with a 95% confidence level.

Table 6: Indicator P-test values

Indicator	C	Water Area	GDP	Club Number	I1	I2	I3
	0.000	0.004	0.000	0.009	0.000	0.741	0.056

The information given by the interaction item shows that the impact of regional economic development on the price of different varieties of sailing ships is consistent. The impact of regional geographical conditions on the price of different varieties of sailing ships is inconsistent, and the impact of regional market demand effect on the price of different varieties of sailing ships is inconsistent.

Relate the multivariate linear regression equation constructed in problem 1 to explain the statistical significance of regional effects.

The regional economic development level effect is exemplified by GDP, where prices increase by 16.6% on average for each trillion dollar increase in GDP on a given state, holding all other variables constant.

The regional geography effect is exemplified by water area. For every 1% increase in water area on a given state, prices fall by an average of 1.2%, holding all other variables constant.

The regional market demand effect is exemplified by club number. For every increase in club in a state, prices increase by 0.02% on average, holding all other variables constant.

Table 7: Model Estimatation Results

Year	Price	GDP(g)	Year	Price	GDP(g)
2005	240.00	1.8520E+13	2009	394.35	3.4563E+13
2006	270.00	2.1766E+13	2010	453.31	4.0890E+13
2007	304.81	2.6802E+13	2011	498.43	4.8412E+13
2008	418.55	3.1675E+13	2012	623.00	5.3412E+13

Aiming at the special economic development factors in the regional effect, the realistic significance of its characteristic index GDP is discussed. The data of China's GDP and China's coal Price from 2005 to 2012 are found to be consistent with the model. In order to avoid misleading conclusions caused by the inversion of causality between GDP and Price, Granger causality test is conducted for $1np_1$ and lng_1 with cointegration relationship, and the final conclusion is drawn: In the long run, there is a causal relationship between coal price and GDP, and the probability of economic growth affecting coal price is higher than the probability of coal price affecting GDP.

5.2 Cross-Introduction Test

In response to the second question, we use another intuitive method to verify the above conclusion: the region has an impact on the listing price. In order to better study the impact of regions on listing prices, we take the United States as an example, divide the regions into whether they are coastal and economically developed, and select Florida, Colorado, New York, and Minnesota as representatives of coastal cities, inland cities, developed cities and economically underdeveloped cities.

When exploring the regional effect of economic development level on prices, taking New York as an example, assuming that the sailing varieties sold in New York City are also sold in Minnesota cities, import these sailing variety parameters and Minnesota regional data into the model to predict the price of these sailing varieties sold in Minnesota. The same is done for Minnesota for the variety of sailboats on sale. We select the mean, sum and maximum values as a reference for the actual and predicted prices to quantitatively study their relationship.

The results show that, in general, the listing price of sailboats for sale in New York is lower if they are sold in Minnesota, and the listing price of sailboats for sale in Minnesota is higher if they are sold in New York. That is, the market price of sailboat is

positively correlated with economic development, and the price of sailboat in developed areas is generally higher than that in underdeveloped areas. But this is not the same in all sailboat variants, some specific sailboat variant's price does not change in the same direction with the level of economic development of the region, which reflects the actual preference of specific types of sailboats in specific regions. It may be related to the local geographical and cultural environment.

In exploring the regional effects of the coast on prices, taking Florida and Colorado as an example. Do the same as above, replace the geographical variables of the sailboat sold here with each other, and observe the relationship between the price changes. The results indicate that the prices of sailboats in coastal areas are generally higher than those in inland areas. But there are still some exceptions. So, the influence of regional factors is not entirely consistent among different sailboat variants.

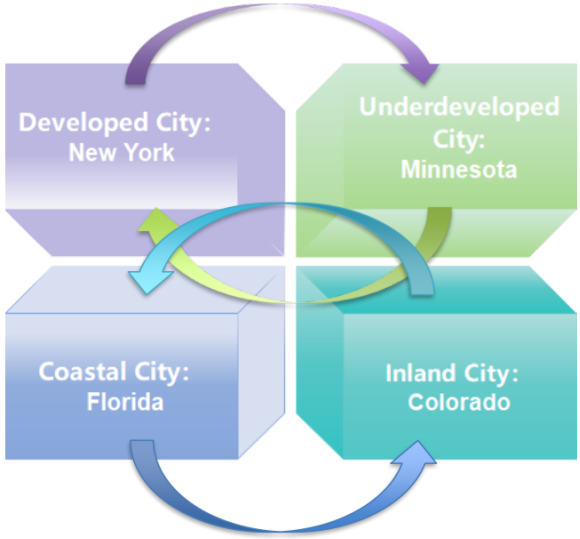


Figure 12: Cross-Introduction Test

6 Judgment Model based on Test of Goodness of Fit

By incorporating regional data from Hong Kong into the multiple linear regression model, the fitted values obtained were analyzed for accuracy with the actual values, with a coincidence of 88.97%. There are sufficient reasons to believe that the impact of the Hong Kong regional effect on Price is consistent with the conclusion reached in question 2.

Then discuss whether the regional effect of Hong Kong (SAR) on catamarans and monohull sailboats is the same. We name the two classification variables of catamarans and monohull sailboat with values of 0 and 1.

The verification procedure is similar to problem 2. First add interaction terms as follow, then constructed t-statistics for consistency tests to obtain the P-test values for each indicator.

$$
\begin{aligned}
J_1 &= Is_Mono * GDP \\
J_2 &= Is_Mono * ClubNumber \\
J_3 &= Is_Mono * WaterArea
\end{aligned} \tag{8}
$$

Analyzing the table below, it can be seen that Water Area, Club Number alone all have an impact on the price with a 95% confidence level. And GDP has an impact on the price with a 90% confidence level. The result show that the effects of regional economic development level, regional geographical conditions and regional market demand on catamarans and monohull sailboats are different.

Table 8: Indicator P-test values

Indicator	Is Mono	Water Area	GDP	Club Number	J1	J2	J3
	0.000	0.004	0.091	0.009	0.269	0.741	0.729

7 Optimization of Model

Optimize random forest model based on hyper parameter optimization.In machine learning, the parameters that can not be optimized by training and need to be selected manually are called hyperparameters. The choice of hyperparameters is very important, and the problem of over-fitting or under-fitting is easy to appear if the choice is not appropriate.

7.1 Selection of hyperparameter

In the random forest regression model, we combined the data situation and decided to tune for several important hyperparameters. They are the number of decision trees, the maximum depth of the decision tree, the number of small separated samples (that is, the minimum number of samples required to split the decision tree node), and the minimum number of leaf node samples (that is, the minimum number of samples that a leaf node needs to contain).

7.2 Randomized search on hyper parameters

First, we determine the value range of each hyperparameter, and use the method of random search matching to arrange and combine them into the model for training. By cross-validation, the results of each superparameter combination are compared, and a set of optimal superparameter combination is determined.

7.3 Grid Search

Hyperparameter random search is a random incomplete traversal process, and the resulting optimal combination may be local optimal rather than global optimal. Therefore, on the basis of the optimal combination obtained by hyperparameter random search, we select the value in its near range, and then use the grid search method to try

each possibility, traverse the value of each hyperparameter, and cross-verify to select the better hyperparameter value.

8 Strengths and Weaknesses

8.1 Strengths

Multiple Linear Regression Model

- It can account for the effects of multiple predictors simultaneously, allowing for a more accurate representation of the complex relationships that influence sailboat pricing.
- It can provide a clear quantification of the relative importance of each predictor in determining sailboat pricing. By calculating regression coefficients and p-values, we can determine which predictors have a significant effect on price, and by how much.

Random Forest Regression Model

- It can handle a large number of predictors, handle non-linear relationships, and reduce the risk of overfitting.
- It can handle both numerical and categorical variables, making them suitable for modeling complex datasets with a large number of predictors.
- It create many decision trees using a subset of the predictors, which helps to reduce the variance and improve the generalizability of the model.

8.2 Weaknessess

Multiple Linear Regression Model

- One limitation is their assumption of a linear relationship between the predictors and the response variable lead model that is too simple and fails to capture important non-linear relationships.
- Another limitation is their sensitivity to outliers and influential observations. The outliers lead to biased predictions.

Random Forest Regression Model

- It's challenging to understand the relationship between the predictors and the response variable and to identify the most important predictors.

9 Used Sailboat List Pricing Explanation Report

First of all, we browsed nearly 10,000 data about the price of second-hand sailboats on numerous second-hand sailboat websites. After learning about the pricing of sailboats, in order to more accurately reflect the impact of regional factors on prices, we found some characteristics, such as GDP, the number of sailing clubs in each region, and so on, as external indicators. Then, we calculated the correlation coefficients between all indices and prices, eliminating indicators with correlation coefficients less than 0.1. At the same time, using feature importance analysis, some noise factors were removed.

Next, we use these screened variables to establish the Multiple Linear Regression Model and the Random Forest Regression Model.

Through our model, we can draw some conclusions: in addition to the inherent attributes of the sailboat, the region is also an important factor affecting the price of the sailboat. Besides, the impact of regional economic level on different types of sailboats is consistent and positive.

Therefore, for Hong Kong, if you want to sell your sailboat at a higher price, you can sell it in an area with a relatively high economic level in Hong Kong

However, it is important to be vigilant against the inconsistency between regional factors and market demand factors with regional effects. We analyzed the data and found that the direction of market activity changes in different regions is not consistent, so we cannot simply apply market reactions from one region to another.

When this phenomenon occurs, we speculate that it may be due to the buyer's personal preference for the product, but we have masked this factor in the model assumptions and did not conduct a specific analysis. This is a flaw in our model, but it is also a direction for improvement in our model.

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